

TEST PROCESS

The following benchmarks were used to evaluate the performance of the IDE hard disk drives in this shootout:

WEIGHTAGE TABLE	
Benchmark	Percent weightage
High-End Winmark 99	10
SiSoft 2001te score	5
SiSoft sequential read	7
SiSoft sequential write	7
SiSoft random read	12
SiSoft random write	7
SiSoft access time	10
HDTach random access time	10
HDTach read burst speed	10
HDTach read speed average	8
HDTach write speed average	8
HDTach CPU utilisation	6

- High-End Disk Winmark 99, which puts the disk sub-system of the PC under stress
 - SiSoft Sandra 2001te (Disk benchmark), a synthetic benchmark that tests the low level parameters of a hard disk such as sequential and random read/write speeds, access times, and so on.
 - HDTach 2.6, which evaluates the sequential read/write, read burst speed, average access time and CPU utilisation of the hard disk.
- After running these tests, the overall score of a hard disk drive was calculated by applying a set of weightages to the various parameters and tests run. The weightages fixed for the tests are as in the table.

Contents

TEST PROCESS107

IDE DRIVES107

7200-rpm drives

IBM Desksta 75GXP 76.8 GB

IBM Deskstar 60 GXP (40 GB)

Samsung SP4004H (40 GB)

Seagate Barracuda ST320414A (20 GB)

5400-rpm drives

Fujitsu MPG3204AT (20 GB)

Fujitsu MPG3409AT (40 GB)

Maxtor 536DX (100 GB)

Maxtor 541DX (20 GB)

Maxtor D540X (80 GB)

Samsung SV4084H (40 GB)

Samsung SV6004H (60 GB)

Seagate ST320410A (20 GB)

Seagate ST340810A (40 GB)

Seagate ST380020A (80 GB)

Western Digital WA Caviar WD200 (20 GB)

BOX

How a Hard Disk Works108

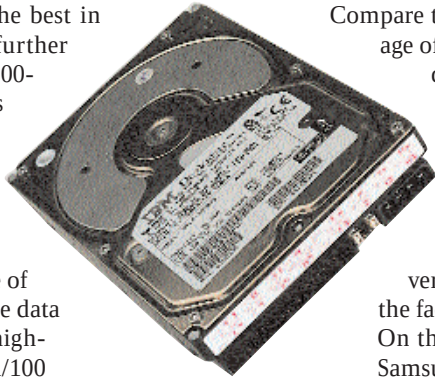
SCORE BOARD111

CONCLUSION113

benchmarking programs that test the performance of each of these sub-systems and also of the hard disk as a whole by using applications such as office productivity suites, etc. The results of these tests are described below.

PERFORMANCE DRIVES (7200-RPM)

When you're looking for the best in IDE disk drives, look no further than the current breed of 7200-rpm speedsters. The platters of these drives spin at a maximum speed of 7200 rpm and by virtue of this, data can be picked up by the read-write heads much faster than in the case of a 5400-rpm drive. Also, since data is read that much faster, high-speed interfaces such as ATA/100 can be properly utilised, which makes the overall performance and response of the hard disk much faster. Let's take a look at this crop of hard disks and see which one is best suited for your applications.



IBM Deskstar 75GXP: this hard drive is silent and has a very low CPU utilisation

Application performance and the Winmark 99

For evaluating the performance of these drives, we used Ziff-Davis High-End Disk Winmark 99. This is a benchmarking tool that uses real-world applications such as office productivity and multimedia authoring tools, comprising audio and video applications that particularly stress the hard disk sub-system of a computer.

These applications are run in a script-driven manner and the index that is returned at the end of this test reflects the drive's performance in working with similar applications. It runs for

about 10 minutes on each hard disk and the strain that the hard disk undergoes is evident from the sound and heat generated during the course of the test.

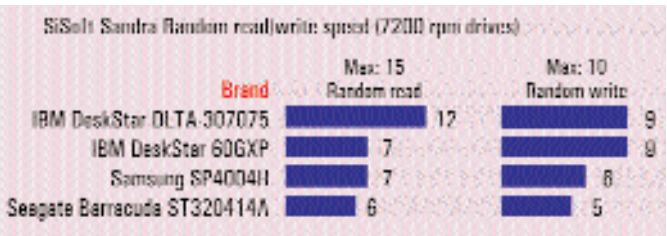
In the 7200-rpm category, the **IBM Deskstar 40 GB** delivered exemplary performance in the Winmark 99 test, logging a score of 26700.

Compare this, for example, to an average of 7500 logged by a 5400-rpm drive of the same capacity.

The other drive that gave a good score in this test was the Seagate Barracuda ST320414A 20 GB hard drive, which scored 23000. This drive was also very economical considering the fact that it's a 7200-rpm drive.

On the other end of the scale, the Samsung SP4004H, with its Spin-Point technology, did not score very well in this test and notched a score of just 5770.

Over to the actual tests that measured the data transfer speeds: SiSoft Sandra 2001te was the program used to evaluate the performance of the drives in these low-level tests. Here, we measured the data transfer speed and the access times for the hard disk. We specifically noted the sequential and random



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More Info on the test process and the products tested.

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